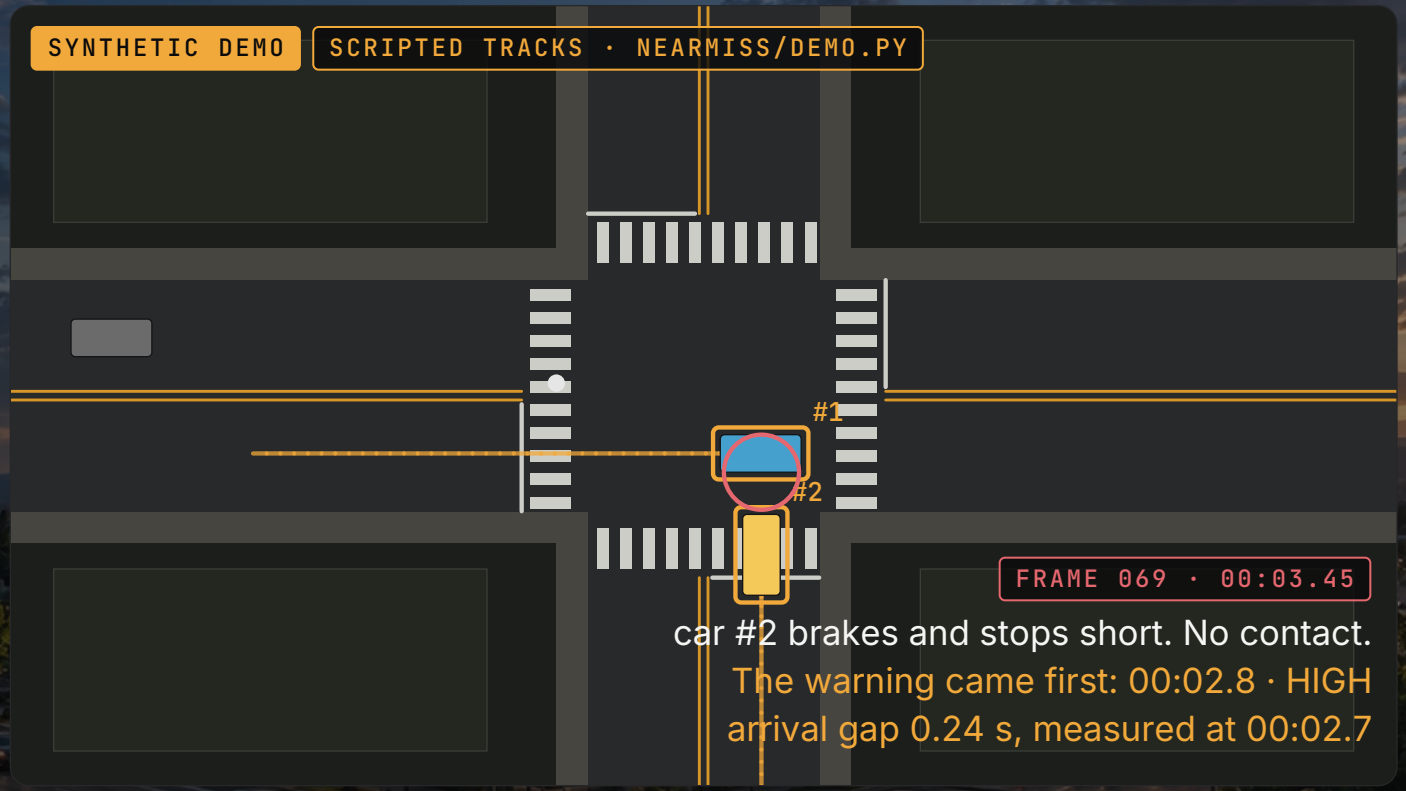


A near miss disappears *in seconds.*

Its evidence shouldn't.

Illustrative image · not analyzed footage

WHAT NEARMISS DOES

Turns fixed-camera traffic footage into *evidence a person can check.*

01
Predict
Projects tracked paths and flags potential conflicts.

02
Detect
Separately screens for contact plus a motion response.

03
Verify
A person reviews the source frame and clip.

04
Notify
Confirmation creates one local, simulated notice.

BUILT BY

Srivibhav Padakandla · Aadi Shukla

WHO IT IS FOR

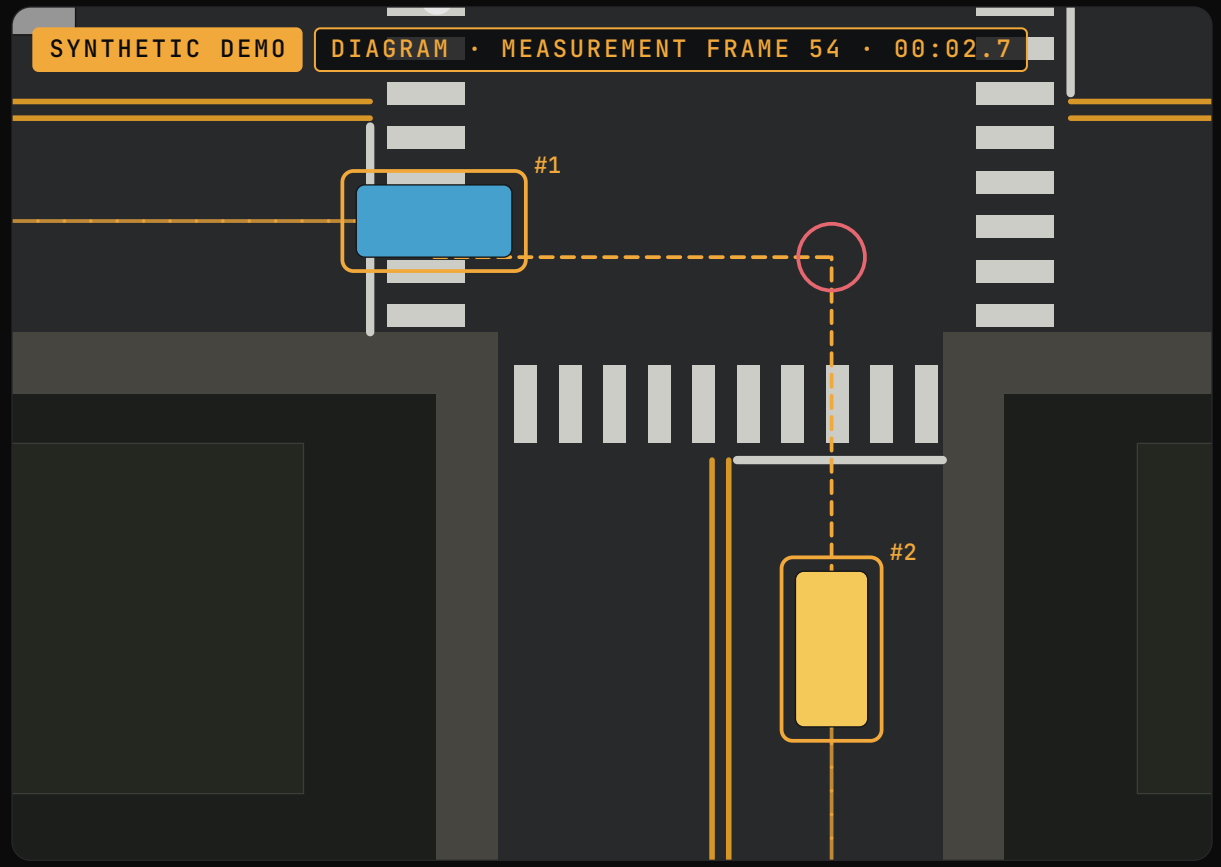
People who review intersections: safety staff, and neighbors preparing a street-safety conversation.

SYNTHETIC DEMO

Every app scene in this talk uses NearMiss's labeled demo: generated footage, scripted tracks. No real place, no real crash.

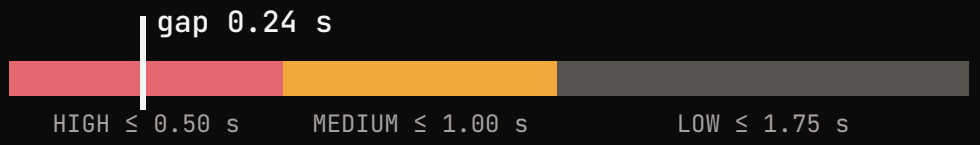
HOW A WARNING IS EARNED

Two projected arrival times. *One gap.*



car #1 reaches the shared point in 0.75 s
car #2 reaches it in 0.99 s
arrival gap 0.24 s → HIGH

Measured at 00:02.7 (frame 54). First alert 00:02.8 (frame 56).



HIGH: two vehicles were projected to reach the same point nearly together

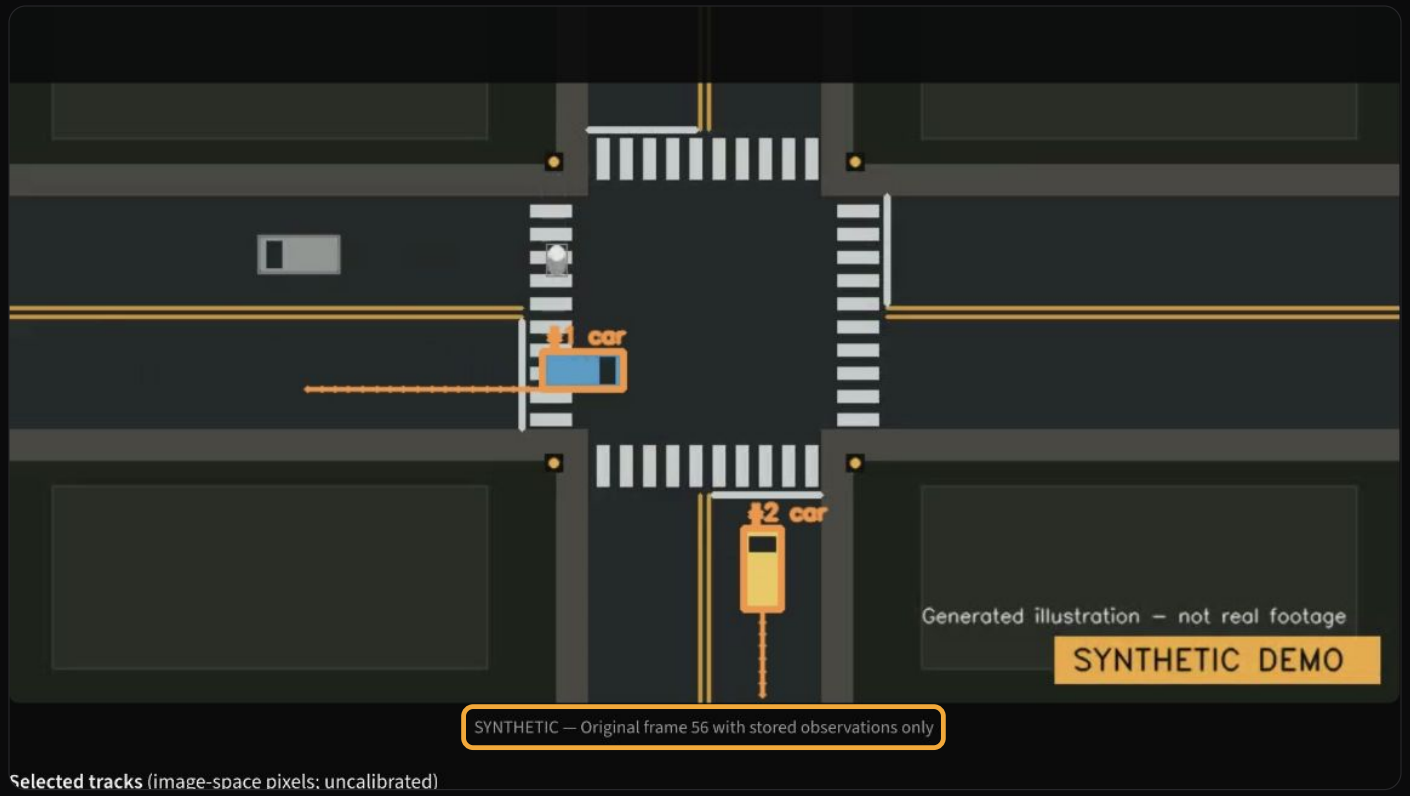
- Projected paths share a conflict point: car #1 would reach it in 0.75s and car #2 in 0.99s.
- Arrival gap is 0.24s (HIGH ≤ 0.50s, MEDIUM ≤ 1.00s, LOW ≤ 1.75s).
- Projected footprints overlap: 0.0px apart against a 16.4px combined footprint.
- The pair was closing at 252px/s in image space.

Actual app · Analysis → “Why each warning was raised”

Image-space pixels and constant-velocity projection. Not a measured speed, distance or collision probability.

INSPECT THE SOURCE

Every flag points back to *the exact frame.*



Actual app · Investigation · frame view with stored observations

00:02.8 HIGH
Crossing paths · car #1 / car #2
arrival gap 0.24 s

00:07.1 HIGH
Crossing paths · car #3 / truck #4
arrival gap 0.03 s

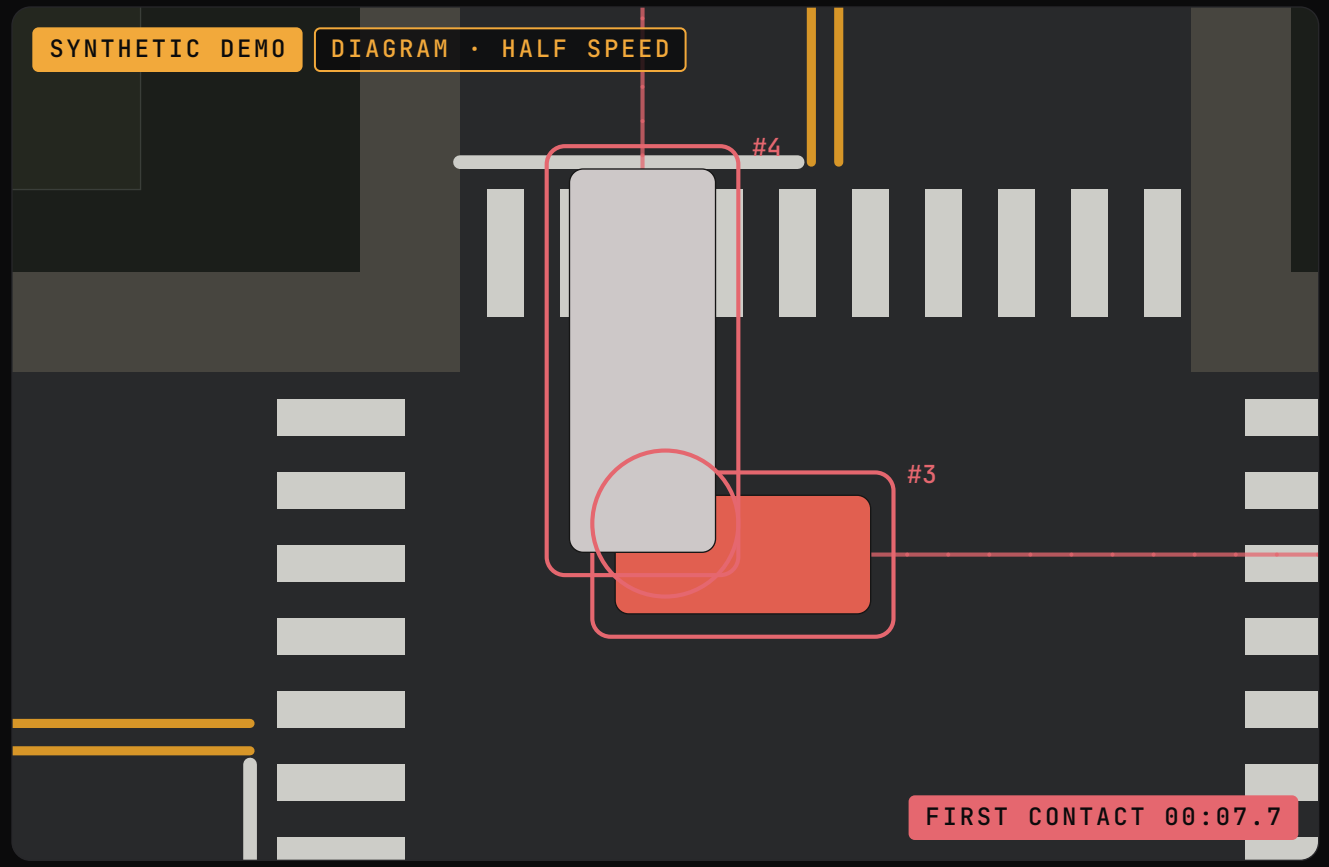
00:07.7 CONTACT
Suspected contact · vehicles #3 / #4
5 supporting signals · needs review

00:09.8 MEDIUM
Same-direction following · car #5 / car #6
time headway 0.48 s

Jump to evidence opens frame 56 (00:02.8), the first alert, with tracks #1 and #2 selected. Boxes are stored observations; nothing is re-detected.

HUMAN REVIEW

Contact needs more than overlap. *A person decides.*



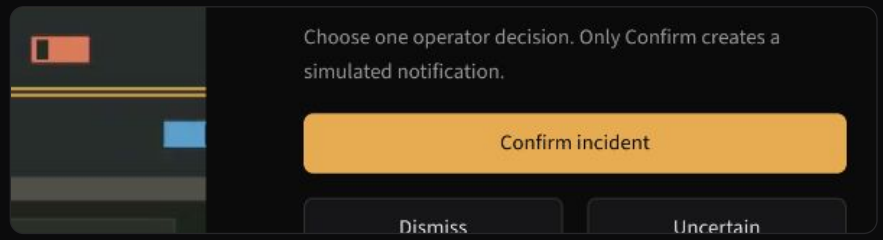
- Persistent box contact
- Significant box overlap
- Converging before contact
- Abrupt motion change
- Sudden deceleration

Prediction conflict-002 came 0.60 s before first contact, but does not decide this. Correlated measurements, not independent votes.

Confirm
creates one local, simulated notice

Dismiss
no notice

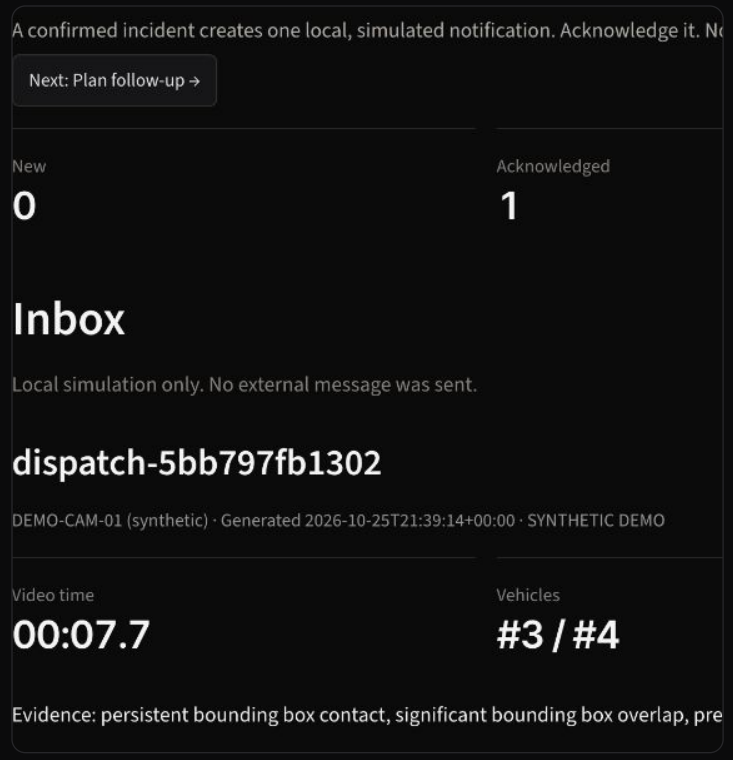
Uncertain
no notice



Actual app · Incidents (review recorded on the synthetic demo)

FROM EVIDENCE TO FOLLOW-UP

One local notice, a plan, and *a brief anyone can check.*



Actual app · Dispatch · local simulation, nothing sent

STREET LAB · CROSSING-TIME ARITHMETIC

Typical walking speed · 3.5 ft/s

19.1 s now

13.4 s hypothetical

Slower walking speed · 2.5 ft/s

26.0 s now

18.0 s hypothetical

60.0 → 40.0 ft crossing · start-up 2.0 s · placeholder assumptions

Arithmetic on entered assumptions. Not a measured crash reduction.

ACTUAL EXPORT · EVIDENCE BRIEF (.MD)

[SYNTHETIC] NearMiss evidence brief: Suspected incident (review candidate)

> SYNTHETIC DEMONSTRATION: generated footage and scripted tracks. Not a real location, real traffic or a real incident.

Human review

An operator marked this as a Confirmed Incident in NearMiss (2026-10-25T21:39:14+00:00). This is a human judgment made from the footage, not an official crash report.

Simulated dispatch

A simulated dispatch notice was created (2026-10-25T21:39:14+00:00) and acknowledged in the local inbox (2026-10-25T21:39:31+00:00). Simulated and local only: NearMiss sent nothing outside this computer.

What this brief does not claim

- It is not proof that a collision or near miss happened; it is computer-vision evidence for human review.
- It contains no crash probability, safety rating, accuracy figure or real-world danger score.
- Distances and speeds are image pixels, not metres or road speeds; no field calibration was applied.

Street Lab's community brief and the Analysis PDF report are separate exports.

WHAT WE BUILT · VERIFIED · HAVE NOT PROVEN

Open-source eyes. *Our own judgment layer. Honest limits.*

BUILT

EXISTING OPEN SOURCE

- Ultralytics YOLO: detects road users
- ByteTrack: keeps IDs across frames
- OpenCV, Streamlit, Plotly, ReportLab

NEARMISS CUSTOM LOGIC

- Trajectory prediction and conflict gates
- Same-direction following check
- Multi-signal suspected-collision detector
- Evidence association and de-duplication
- Review record and simulated dispatch audit trail
- Investigation, briefs, Street Lab, run library

VERIFIED

- Automated tests: 237 passed (local pytest, 2026-10-25)
- Deterministic demo: ID synthetic-demo-3224408e7e74; 3 warnings and 1 suspected incident every run at default settings
- Annotated clip regenerated for this deck matches the app’s cached render byte for byte
- Synthetic label carried through frames, pages and exports
- Confirm, dismiss and acknowledge states covered by the tests (SQLite review record)

NOT YET PROVEN

- Accuracy on real-world footage has not been measured
- Image pixels only: no calibrated metres, speeds or probabilities
- No crash-reduction or safety-rating claim
- Dispatch is simulated; nothing leaves the computer
- One camera view, short clips

PROPOSED NEXT STEP

A supervised pilot *before any claims.*

01

Permissioned footage from one fixed camera

02

Compare NearMiss flags with human-labelled events to measure accuracy

03

Calibrate the camera view so pixels can become metres

AI ASSISTANCE

OpenAI Codex and Anthropic Claude (Opus) assisted with writing, reviewing and testing code, and with building this deck. Results shown come from running the app and its tests.

OPEN SOURCE

Ultralytics YOLO · ByteTrack · OpenCV · Streamlit · Plotly · ReportLab · GSAP · Inter, Instrument Serif, JetBrains Mono (OFL)

A near miss disappears *in seconds. Its evidence shouldn't.*

NearMiss · Srivibhav Padakandla · Aadi Shukla

Illustrative image · not analyzed footage